

Sam Freedman

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Citizenship: United States

Research Interests

Teichmüller dynamics; translation surfaces; computational methods in geometry and dynamics

Employment

University of Chicago, Chicago, IL

Sept 2024–present

L.E. Dickson Instructor, Department of Mathematics

Susquehanna International Group (SIG), Bala Cynwyd, PA

June–Aug 2026

Quantitative Research and Systematic Trading Intern

- Built and evaluated predictive models for trade selection using Python; analyzed model signals and performance using market data.

Education

Brown University

Providence, RI

Ph.D. in Mathematics, May 2024 Advisor: Jeremy Kahn

Sc.M. in Mathematics, May 2020

University of Michigan

Ann Arbor, MI

B.S. in Mathematics, High Honors, May 2018 Minor in Computer Science

Publications

1. S. Freedman and B. Zykoski, “Convex iso-Delaunay regions in strata of translation surfaces,” to appear in *Israel J. Math.* (2026). [arXiv:2509.19550](#)
2. T. Lucas and S. Freedman, “Veech fibrations,” to appear in *Algebr. Geom. Topol.* (2025). [arXiv:2310.02325](#)
3. J. Boulanger and S. Freedman, “There are no primitive Teichmüller curves in $\text{Prym}(2,2)$,” *C. R. Math.* 362 (2024). [doi:10.5802/crmath.551](#)
4. S. Freedman, “Periodic points of Prym eigenforms,” *J. Mod. Dyn.* 20 (2024). [doi:10.3934/jmd.2024004](#)
5. Z. Chowdhury, S. Freedman, *et al.*, “Computing periodic points on Veech surfaces,” *Geom. Dedicata* 217 (2023). [doi:10.1007/s10711-023-00804-z](#)
6. J. Hlavinka, S. Kannan, and S. Freedman, “Automorphisms of tropical Hassett spaces,” *Port. Math.* 79 (2022). [doi:10.4171/pm/2075](#)

Selected Talks

- AMS Eastern Sectional, Boston College, Mar 2026
- Geometry Seminar, Indiana University, Dec 2025
- AMS Central Sectional, Saint Louis University, Oct 2025
- Dynamics Seminar, Northwestern University, Feb 2025
- Dynamics Seminar, University of Bristol, Sept 2024
- Geometry/Topology Seminar, CUNY, May 2024
- Geometry/Topology Seminar, Indiana University, Apr 2024
- Group Actions Seminar, UCSD, Mar 2024
- Geometry/Topology Seminar, Yale, Feb 2024

Teaching

University of Chicago

- Basic Algebra I–II (Math 254–255), Autumn 2026–Winter 2027, Instructor (scheduled)
- Honors Calculus I–II (Math 161–162), Autumn 2025–Winter 2026, Instructor
- Honors Calculus I–III (Math 161–163), Autumn 2024–Spring 2025, Instructor
- Mathematical Methods in the Physical Sciences (Math 184), Winter 2025, Instructor

Brown University

- Instructor/TA II and TA, 2019–2024: Math 520 (Spring 2024, Spring 2023); Math 200 (Fall 2023, Fall 2022); Math 190 (Fall 2021); Math 100 (Winter 2021, Fall 2019); Math 90 (Winter 2020).

Honors and Grants

- Excellence in Teaching Award, Brown University Math Dept., Aug 2022
- Junior Scientific Visibility Program Grant, Fondation Hadamard, Mar 2022

Service and Mentoring

- Mentor, Morehouse/Spelman DRP, Fall 2024
- Organizer, Brown Horizons Seminar, 2023–2024
- Organizer & SAO Liaison, Brown Math Circle, 2019–2024
- Directed Reading Program mentor, Brown University, 2020–2022
- BUMP co-organizer and mentor (Group 5), Brown University, Summer 2020

Professional Activities

- Refereeing: *Mathematische Zeitschrift*; *Discrete and Continuous Dynamical Systems*; *IMRN*